

Haotian Chen (Chahot)

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in LinkedIn

🌐 Homepage

📖 Google Scholar

📝 Technical Blog (in Chinese)



Research Interests

- Questions 📌 My current research interests lie in the theoretical foundations of quantum computing, especially in how quantum algorithms exploit algebraic, spectral, and Fourier-analytic structures. I am particularly interested in whether quantum access models, such as quantum examples and Fourier sampling, can provide provable advantages for testing structural properties of Boolean functions, and how these advantages can be characterized through sample complexity and computational complexity.
- Directions 📌 My current directions include: (1) Quantum Algorithms; (2) Fourier Analysis of Boolean Functions; (3) Property Testing and Spectral Testing; (4) Computational Complexity; and (5) Quantum Information and Cryptography.

Experience

- Oct. 2024 — Apr. 2025 📌 **National University of Singapore**, Singapore
Research Assistant, Centre for Quantum Technologies
Worked on quantum authentication and capability certification under adversarial settings, with attention to proof-of-quantumness-style protocols. Studied unitary inversion and trap-based verification ideas for distinguishing honest and malicious quantum behaviours.
- Sep. 2021 — Oct. 2023 📌 **Seoul National University of Science and Technology**, Seoul, South Korea
Ph.D. Scholar, Department of Computer Science and Engineering
Built research foundations in cybersecurity, cryptographic systems, and distributed computing, with work spanning IoT security, blockchain, digital twins. Later redirected my research trajectory toward quantum computing and theoretical foundations of security.
- Dec. 2023 — Jul. 2024 📌 **Publishing House of Electronics Industry**, Beijing, China
Independent Author
Authored a Chinese-language technical textbook on quantum security, covering classical-to-quantum cryptography, quantum information-theoretic security, QKD, and post-quantum cryptography.
- Feb. 2022 — Nov. 2023 📌 **Seoul National University of Science and Technology**, Seoul, South Korea
Assistant Lab Manager, Ubiquitous Computing and Security Lab
Supported grant proposal preparation, project coordination, academic administration, and lab-level research collaboration.

Experience (continued)

Mar. 2026 — Present

- **Invisible Technologies**, Remote and part-time **AI Trainer (Project-Based)**
Evaluated Korean- and English-language LLM outputs for quality, linguistic accuracy, and instruction adherence.

Education

Sep. 2021 – Nov. 2023

- **Ph.D.(Incomplete), Seoul National University of Science and Technology**. Department of Computer Science and Engineering.
Specialisation: Cybersecurity in IoT
Thesis title: *A Smart Contract and Digital Twin-based Heuristic Multi-Cooperation Scheduling Framework for Smart Manufacturing in IIoT Environments*.
GPA: 4.27/4.50 (97.4/100)
Programme concluded after a strategic transition of research focus to quantum computing.

Mar. 2017 – Aug. 2021

- **Bachelor of Science in Computer Science and Engineering, Seoul National University of Science and Technology**. Department of Computer Science and Engineering.
Specialisation: Information Security
Thesis title: *Servers—Client Mini-Network Security Prototype Based on VMWare and GNS3*.
GPA: 3.64/4.50 (90.2/100)
Rank: Ranked **10th** among Department of Computer Science and Engineering graduates in the Class of 2021.

Teaching & Academic Services

Mar. 2022 - Sep. 2023

- **Teaching Assistant** at Seoul National University of Science and Technology
Assisted in course administration, student coordination, and exam logistics. Responsible for designing exam questions, grading exams and assignments, and maintaining academic records.

Sep. 2019 - Dec. 2020

- **Physics Mentor** at Seoul National University of Science and Technology.
Provided one-on-one tutoring to undergraduate students, assisting them in mastering fundamental physics concepts and problem-solving techniques.

Skills

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|--------------------|---|
| Languages | ■ Chinese (Native); English (IELTS 7.0); Korean (TOPIK Lv.6). |
| Coding | ■ Python, Qiskit, Java, C++, $\LaTeX$. |
| Software and Tools | ■ Docker, Wireshark, Nmap, Jupyter Notebooks, GNS3, VMware, OpenAI API, Scopus, Web of Science, AI-assisted research tools. |
| Miscellaneous | ■ Strategy-based games, fitness and swimming, technical blogging, intercultural communication |

Recognitions

Scholarships

- 2017 — 2021  **Scholarship for Outstanding Achievement for Foreigners** for each semester of my undergraduate studies.
- 2021 — 2023  **Scholarship for Outstanding Achievement for Foreigners** for each semester of my graduate studies.

Best Paper Award(Multiple)

- 2022  The 2022 World Congress on Information Technology Applications and Services Jeju, Korea
-  The International Conference on Big data, IoT, and Cloud Computing, Jeju, Korea
-  The International Conference on Future Information Technology, applications and services, Seoul, Korea
-  The International Conference on Computer Science and its Applications, Vientiane, Laos

Certification and Professional Service

- 2023  **Alibaba Cloud Developer Community-Blogging Expert.** Alibaba Changfengzhe Program.
-  **CSDN-Blogging Expert.** Subscriptions more than 20,000 and total number of visits is about 600,000.
-  **Examiner.** Working for the security section of the Korean Information Processing Technology Professional Certification Examination.

Research Publications

Journal Articles

- 1 **H. Chen**, A. E. Azzaoui, H. Park, D. Camacho, and J. H. Park, "A comprehensive study of quantum computing technologies in smart city: Review and future directions," *Human-centric Computing and Information Sciences*, vol. 14, 2024, Impact Factor: 3.9, JCR Q2 Top 26.3%.
- 2 S. Jeremiah, **H. Chen**, S. Gritzalis, and J. H. Park, "Leveraging application permissions and network traffic attributes for android ransomware detection," *Journal of Network and Computer Applications*, vol. 230, p. 103 950, 2024, Impact Factor: 7.7, JCR Q1 Top 4.2%.
- 3 **H. Chen**, A. Azzaoui, S. Jeremiah, and J. H. Park, "A novel smart contract based optimized cloud selection framework for efficient multi-party computation," *Journal of Information Processing Systems*, vol. 19, no. 2, pp. 240–257, 2023, Scopus.
- 4 **H. Chen**, S. Jeremiah, C. Lee, and J. H. Park, "A digital twin-based heuristic multi-cooperation scheduling framework for smart manufacturing in iiot environment," *Applied Sciences*, vol. 13, no. 3, p. 1440, 2023, Impact Factor: 2.5, JCR Q2 Top 24.3%.
- 5 A. Azzaoui, **H. Chen**, S. Kim, Y. Pan, and J. H. Park, "Blockchain-based distributed information hiding framework for data privacy preserving in medical supply chain systems," *Sensors*, vol. 22, no. 4, p. 1371, 2022, Impact Factor: 3.4, JCR Q2 Top 30.9%.

Conference Proceedings

- 1 **H. Chen** and J. H. Park, "Quantum-powered secure multi-party collaboration framework for smart manufacturing," in *Proceedings of the International Conference on Big Data, IoT, and Cloud Computing (BIC)*, Jeju, Korea, Aug. 2023.
- 2 A. E. Azzaoui, **H. Chen**, and J. H. Park, "Qaoa: A quantum approximate optimization algorithm for optimal base station selection in 6g networks and beyond," in *Proceedings of the International Conference on Computer Science and its Applications (CSA)*, National University of Laos, Laos, Dec. 2022.
- 3 **H. Chen**, T. W. Kim, and J. H. Park, "Smart contract-based secure multi-party computation for cloud privacy preserving," in *Proceedings of the World Congress on Information Technology Applications and Services (World IT Congress)*, Jeju, Korea, Feb. 2022.
- 4 **H. Chen**, T. Kim, J. Park, and J. H. Park, "Optimized node selection method for efficient cloud computing based on smart contracts," in *Proceedings of the Korean Information Processing Society Conference*, vol. 29, 2022, pp. 48–51.
- 5 **H. Chen** and J. H. Park, "Bdt: Blockchain and digital twins-based secure framework for efficient data processing in smart manufacturing," in *Proceedings of the International Conference on Big Data, IoT, and Cloud Computing (BIC)*, Jeju, Korea, Aug. 2022.
- 6 **H. Chen** and J. H. Park, "Quantum homomorphic cryptography-based efficient medical imaging architecture for iomt in smart city environments," in *Proceedings of the International Conference on Future Information Technology, Applications and Services (IFIT)*, Seoul, Korea, Oct. 2022.
- 7 **H. Chen**, T. Kim, J. Park, and J. H. Park, "Secure multi-party computation based on homomorphic encryption for privacy preserving in iot networks," in *Proceedings of the Korea Information Processing Society Conference*, 2021, pp. 189–192.
- 8 **H. Chen**, T. Kim, and J. H. Park, "Multi-blockchain system based on directed acyclic graph for improving data processing performance," in *Proceedings of the Korean Information Processing Society Conference*, vol. 28, 2021, pp. 25–28.

Books

- 1 Chahot, *Quantum Security: A New Era in Information Protection*, Chinese. Beijing, China: Publishing House of Electronics Industry, 2024, A Chinese-language technical textbook providing a systematic overview of classical-to-quantum cryptography, quantum key distribution, and the transition from classical to quantum-based security frameworks., ISBN: 978-7-121-48136-9.  URL: https://www.phei.com.cn/module/goods/wssd_content.jsp?bookid=65109.

References

	Academic Mentor-Prof. Laurence T. Yang  ltyang@zzu.edu.cn FCAE, FEIC, MAE, MRAE, MNAAI, FIEEE Academic Vice-President Zhengzhou University, China
	Course Lecturer-Prof. Dongwann Kang  dongwann@seoultech.ac.kr Associate Professor, Department of Computer Science and Engineering Seoul National University of Science and Technology, South Korea
	Academic Instructor-Prof. Greg Thompson  gthompson@kaist.ac.kr Adjunct Professor, KAIST Language Center Korea Advanced Institute of Science and Technology, South Korea